

8.3.3 Conjugate Gradient Method Basics ¶

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fit width



The idea behind the Conjugate Gradient Method is that in the current iteration we have an approximation, $x^{(k)}$ to the solution to $Ax = b$. By construction, since $x^{(0)} = 0$,

$$x^{(k)} = \alpha_0 p^{(0)} + \dots + \alpha_{k-1} p^{(k-1)}.$$

Also, the residual

$$\begin{aligned} r^{(k)} &= \\ &= b - Ax^{(k)} \\ &= b - A(\alpha_0 p^{(0)} + \dots + \alpha_{k-1} p^{(k-1)}) \\ &= b - \alpha_0 Ap^{(0)} - \dots - \alpha_{k-1} Ap^{(k-1)} \\ &= r^{(k-1)} - \alpha_{k-1} Ap^{(k-1)}. \end{aligned}$$

If $r^{(k)} = 0$, then we know that $x^{(k)}$ solves $Ax = b$, and we are done.

Assume that $r^{(k)} \neq 0$. The question now is "How should we construct a new $p^{(k)}$ that is A-conjugate to the previous search directions and so that $p^{(k)T} r^{(k)} \neq 0$?" Here are some thoughts:

- We like the direction of steepest descent, $r^{(k)} = b - Ax^{(k)}$, because it is the direction in which $f(x)$ decreases most quickly.
- Let us chose $p^{(k)}$ to be the vector that is A-conjugate to $p^{(0)}, \dots, p^{(k-1)}$ and closest to the direction of steepest descent, $r^{(k)}$:

$$\|p^{(k)} - r^{(k)}\|_2 = \min_{p \perp \text{Span}(Ap^{(0)}, \dots, Ap^{(k-1)})} \|r^{(k)} - p\|_2.$$

This yields the algorithm in [Figure 8.3.3.1](#).

```
Given : A, b
x^(0) := 0
r^(0) := b
k := 0
while r^(k) != 0
  if k = 0
    p^(k) = r^(0)
  else
    p^(k) minimizes min_{p \perp Span(Ap^(0), ..., Ap^(k-1))} ||r^(k) - p||_2
  endif
  alpha_k := (p^(k)T r^(k)) / (p^(k)T Ap^(k))
  x^(k+1) := x^(k) + alpha_k p^(k)
  r^(k+1) := r^(k) - alpha_k Ap^(k)
  k := k + 1
endwhile
```

Figure 8.3.3.1. Basic Conjugate Gradient Method.